(a) Latent Visual Reasoning Inference

[Answer]

RIGHT, UP, UP, RIGHT

lm head

Multimodal LLM f_θ

<|latent_start|>

<|latent_end|>

[Image]

" Here is the map Please provide your action plan. "

[Text]

image embedding

text embedding

special token

latent embedding

(b) Warm Up Alignment

f_{enc}

Projector

MeanPool

Cosine Similarity

MLLM

latent hidden state

h_1

h_2

Ans

CE Loss

(d) Trajectory Contrastive Reward

Rollout Trajectory

Tr_{pos} Correct Answer

Tr_{neg} Incorrect Answer

Tr_{now} Update Trajectory

pull

push

Format Reward

Trajectory-level Contrastive Reward

Correct Reward

$$R_{total} = R_{format} + R_{traj-contras} + R_{correct}$$

(c) Contrastive Latent Supervision

f_{enc}

proj

S_I

$S_{I_{hint}}$

Neg

Pos

Δ

S^j_{neg}

MLLM

latent hidden state

h_i

$\mathcal{L}_{InfoNCE} + \mathcal{L}_{CE}$